

Laws of Indices — Paper D

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is small enough to handle on paper, and every fraction comes out simple.
- The laws are not printed on this paper. You are expected to know them.
- The mark for each question is shown in square brackets on the right.
- Several questions describe steps you may not have seen before — apply the ideas you already have. If two bases do not match, ask whether one of them can be rewritten in terms of the other.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – POWERS OF TEN AND PLACE VALUE

1. Complete the table. Each empty box is worth one mark. [4]

Number	As a power of 10
1000	
1 000 000	
0.01	
0.0001	

For the last two, write the number as a fraction first. One over a power of ten is a negative index.

2. Work with base 10 throughout.

(a) Write $10^4 \times 10^3$ as a single power of 10. [1]

(b) Write $10^5 \div 10^8$ as a single power of 10. [1]

(c) Write your answer to part (b) as a decimal. [1]

3. A sheet of paper is about 10^{-4} m thick. A stack contains 10^3 sheets.

(a) Write the thickness of the stack as a single power of 10, in metres. [2]

(b) Write that thickness as an ordinary number, in metres and in centimetres. [1]

(c) A student writes " $10^{-4} \times 10^3 = 10^{-12}$ ". Explain the mistake in one sentence. [1]

SECTION 2 – WHY THE LAWS HAVE TO BE TRUE

4. This question is about the zero index.

(a) Use $6^4 \div 6^4$ and the division law to show that $6^0 = 1$. [2]

(b) Explain why the same argument works whatever base you choose. [1]

(c) Explain what goes wrong with the argument when the base is 0. [1]

5. This question is about negative indices.

(a) Use the division law to write $5^2 \div 5^5$ as a single power of 5. [1]

(b) Write both powers out in full and cancel, to find the value as a fraction. [2]

(c) Use your two answers to explain what a negative index means. [1]

6. Write these four values in order, smallest first. Show the value of each one as a fraction. [3]

$$10^{-2} \quad 2^{-3} \quad 5^{-1} \quad 3^{-2}$$

All four are one over something. The bigger the bottom, the smaller the fraction.

SECTION 3 – CORRECTIONS

7. Simplify fully.

(a) $(3x^2)^3 \div 9x^4$ [2]

(b) $(a^5 \times a^{-2})^2$ [2]

8. A student has written two lines in her book, and both are wrong.

[4]

(i) " $(x^3)^2 = x^5$ "

(ii) " $10^{-2} = -100$ "

For each one, explain the mistake and give the correct answer.

End of paper. Check every answer has a unit or a form the question actually asked for.